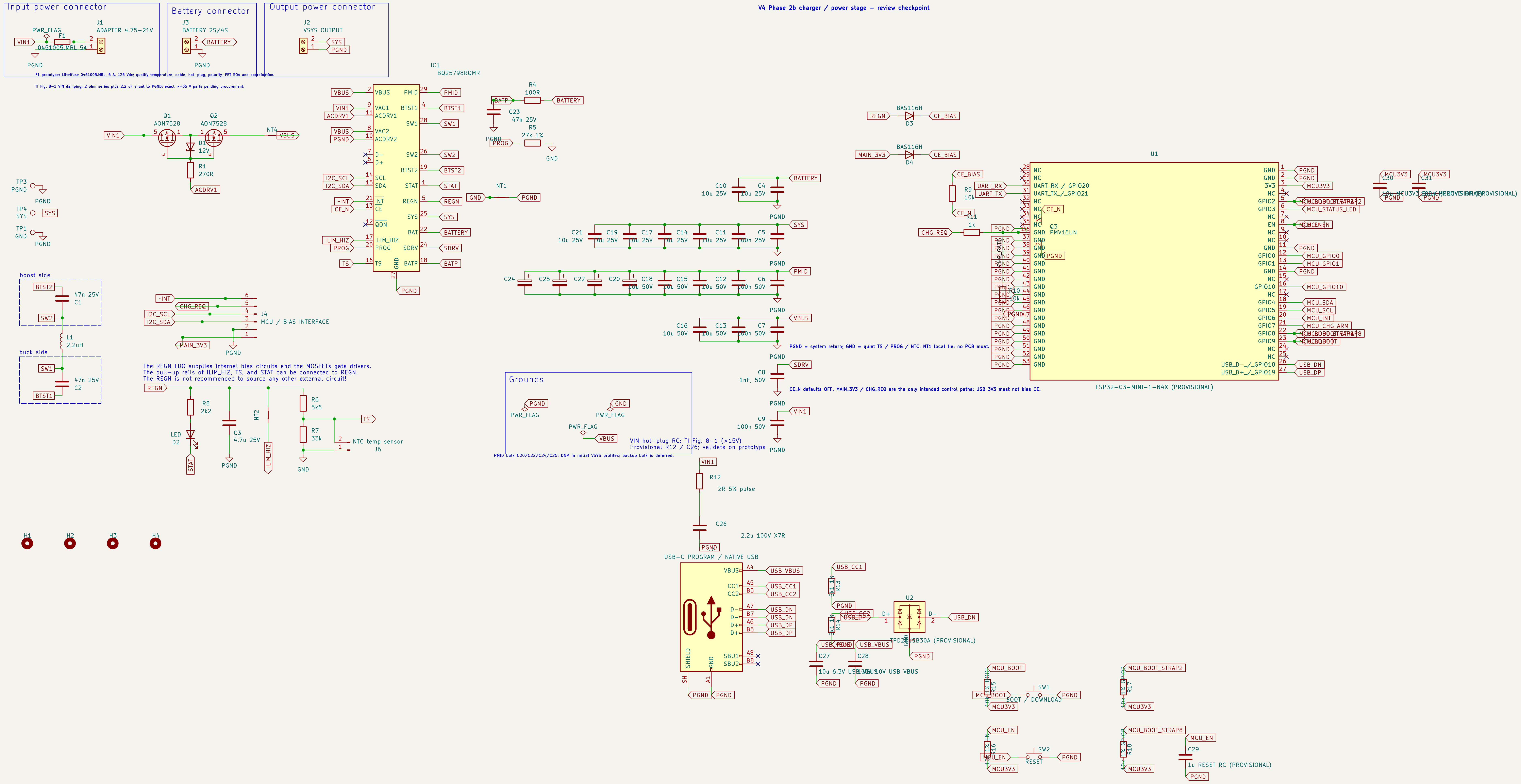
V4 Phase 2b charger / power stage – review checkpoint


Input power connector
PWR_FLAG
F1
0451005.MRL 5A
J1
ADAPTER 4.75–21V
VIN1
PGND
Battery connector
J3
BATTERY 2S/4S
BATTERY
Output power connector
J2
VSYS OUTPUT
SYS
IC1
BQ25798RQMR
VBUS
VAC1
ACDRV1
VAC2
ACDRV2
D–
D+
SCL
SDA
INT
CE
QON
ILIM_HIZ
PROG
TS
GND
PMID
BTST1
SW1
SW2
BTST2
STAT
REGN
SYS
BAT
SDRV
BATP
Q1
AON7528
Q2
AON7528
D1
12V
R1
270R
NT4
R4
100R
C23
47n 25V
R5
27k 1%
NT1
TP3
TP4
TP1
boost side
buck side
BTST2
47n 25V
C1
SW2
L1
2.2uH
SW1
C2
BTST1
J4
MCU / BIAS INTERFACE
~INT
CHG_REQ
I2C_SCL
I2C_SDA
MAIN_3V3
REGN
R8
2k2
LED
D2
C3
4.7u 25V
NT2
R6
5k6
R7
33k
NTC temp sensor
J6
Grounds
PWR_FLAG
C10
10u 25V
C4
10u 25V
C21
10u 25V
C19
C17
C14
C11
C5
100n 25V
C24
C25
C22
C20
C18
10u 50V
C15
C12
C6
100n 50V
C16
C13
C7
C8
1nF, 50V
SDRV
C9
100n 50V
R12
2R 5% pulse
C26
2.2u 100V X7R
BAS116H
D3
D4
CE_BIAS
R9
10k
R11
1k
Q3
PMV16UN
U1
ESP32–C3–MINI–1–N4X (PROVISIONAL)
UART_RX_/_GPIO20
UART_TX_/_GPIO21
3V3
EN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
USB_D–_/_GPIO18
USB_D+_/_GPIO19
MCU3V3
MCU_STATUS_LED
MCU_GPIO0
MCU_GPIO1
MCU_GPIO10
MCU_SDA
MCU_SCL
MCU_INT
MCU_CHG_ARM
MCU_BOOT
MCU_BOOT_STRAP2
MCU_BOOT_STRAP8
MCU_EN
R10
10k
C30
C31
100n MCU3V3 DECOUPLING (PROVISIONAL)
USB–C PROGRAM / NATIVE USB
VBUS
CC1
CC2
D–
D+
SBU1
SBU2
SHIELD
SH
A1
USB_VBUS
USB_CC1
USB_CC2
USB_DN
USB_DP
R13
R14
U2
TPD2E2U06 USB30A (PROVISIONAL)
C27
10u 6.3V USB VBUS
C28
100n 10V USB VBUS
SW1
BOOT
DOWNLOAD
SW2
RESET
R15
R16
R17
R18
10k 1%
C29
1u RESET RC (PROVISIONAL)
H1
H2
H3
H4

F1 prototype: Littelfuse 0451005.MRL, 5 A, 125 Vdc; qualify temperature, cable, hot-plug, polarity-FET SOA and coordination.

TI Fig. 8-1 VIN damping: 2 ohm series plus 2.2 uF shunt to PGND; exact >=35 V parts pending procurement.

The REGN LDO supplies internal bias circuits and the MOSFETs gate drivers.
The pull-up rails of ILIM_HIZ, TS, and STAT can be connected to REGN.
The REGN is not recommended to source any other external circuit!

PGND = system return; GND = quiet TS / PROG / NTC; NT1 local tie; no PCB moat.

CE_N defaults OFF. MAIN_3V3 / CHG_REQ are the only intended control paths; USB 3V3 must not bias CE.

VIN hot-plug RC: TI Fig. 8-1 (>15V)
Provisional R12 / C26; validate on prototype

PMID bulk C20/C22/C24/C25: DNP in initial VSYS profiles; backup bulk is deferred.

PHASE 3.1 CHECKPOINT: native USB / recovery wiring only

USB_VBUS is USB-only input; USB-derived 3V3 regulator is pending next checkpoint.

MCU3V3 is a reserved power-domain input; VSYS buck, USB buck and mux are pending next checkpoint.

PCB BLOCK: exact Espressif land-pattern comparison pending; readback is 62 physical pads / 60 unique numbers, duplicate ground numbers, no courtyard.

Sheet: /
File: ups12v_v4.kicad_sch
Title: BQ25798 UPS module
Size: A2 Date: **Rev: 2.1**
KiCad E.D.A. 10.0.7 Id: 1/1